Key features of parabolas

1

- **a** 5 units
- **b** 5 units
- **c** 10 units
- **d** 10 units

2

- **a** 5 units
- **b** 5 units
- **c** 10 units
- **d** 10 units

3

a

- **i** Upwards
- **ii** $(2, 4)$
- **iii** $a = 3$
- **iv** $(2, 7)$
- **v** $y = 1$

b

- **i** Upwards
- **ii** $(-4, -2)$
- **iii** $a = \frac{3}{2}$
- **iv** $\left(-4, -\frac{1}{2}\right)$
- **v** $y = -\frac{7}{2}$

c

- **i** Downwards
- **ii** $(3, -2)$
- **iii** $a = 4$
- **iv** $(3, -6)$
- **v** $y = 2$

d

- **i** Upwards
- **ii** $(0, 3)$
- **iii** $a = \frac{1}{4}$
- **iv** $\left(0, \frac{13}{4}\right)$
- **v** $y = \frac{11}{4}$

e

- **i** Upwards
- **ii** $(0, 0)$
- **iii** $a = 4$
- **iv** $(0, 4)$
- **v** $y = -4$

f

- **i** Upwards
- **ii** $(-2, 0)$
- **iii** $a = 4$
- **iv** $(-2, 4)$
- **v** $y = -4$

4

a

i To the right
ii $(0, 3)$
iii $a = 3$
iv $(3, 3)$
v $x = -3$

b

i To the right
ii $(2, 0)$
iii $a = 4$
iv $(6, 0)$
v $x = -2$

c

i To the right
ii $(2, 3)$
iii $a = 4$
iv $(6, 3)$
v $x = -2$

d

i To the left
ii $(2, -4)$
iii $a = 3$
iv $(-1, -4)$
v $x = 5$

5

a

i $(0, 0)$
ii $x = 0$
iii $a = 4$
iv $(0, 4)$
v $y = -4$

b

i $(0, 0)$
ii $x = 0$
iii $a = \frac{1}{12}$
iv $\left(0, -\frac{1}{12}\right)$
v $y = \frac{1}{12}$

c

i $(0, 0)$
ii $y = 0$
iii $a = 5$
iv $(-5, 0)$
v $x = 5$

d

i $(0, 0)$
ii $y = 0$
iii $a = \frac{1}{20}$
iv $\left(-\frac{1}{20}, 0\right)$
v $x = \frac{1}{20}$

6

a $(2, -1)$
b $(2, 1)$
c $y = -3$
d $x = 2$
e $(-\infty, \infty)$
f $[-1, \infty)$

7

a $(-1, 3)$

b $(1, 3)$

c $x = -3$

d $y = 3$

e $[-1, \infty)$

f $(-\infty, \infty)$

8

a $x = \pm 2\sqrt{6}$

b $\left(2\sqrt{6}, -7\right)$ and $\left(-2\sqrt{6}, -7\right)$

9

a $y = \pm 2\sqrt{15}$

b $\left(-7, 2\sqrt{15}\right)$ and $\left(-7, -2\sqrt{15}\right)$

10 $k = -2, -6$

Graphs of parabolas

11

a Downwards

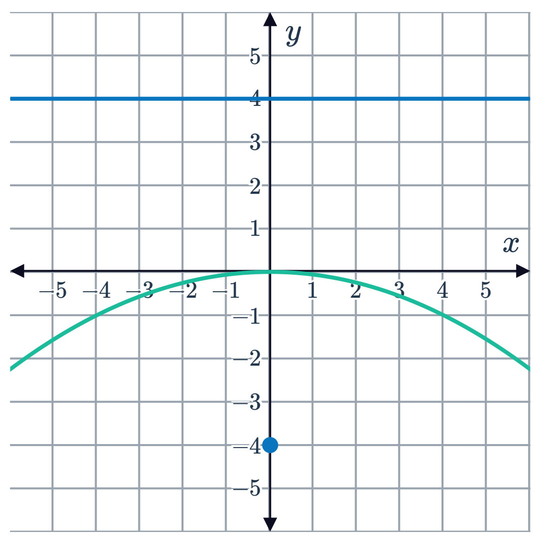
b $(0, 0)$

c $a = 4$

d $(0, -4)$

e $y = 4$

f

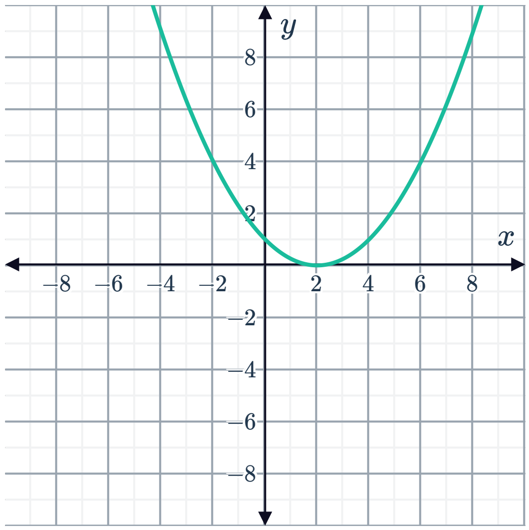


12

a $(0, 1)$

b $y = -1$

c



d $(x-2)^2 = 4y$

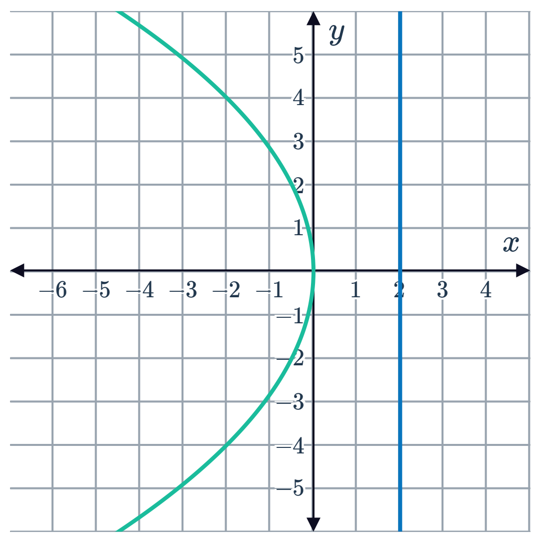
e $(2, 1)$

f $y = -1$

13

a $x = 2$

b



c $y = \pm 4$

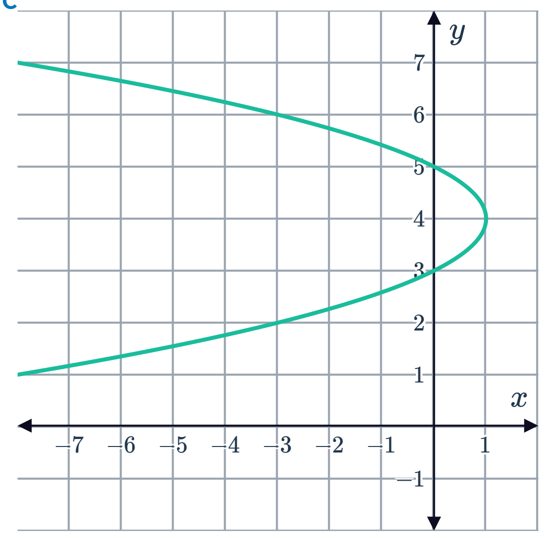
d $(-2, 4)$ and $(-2, -4)$

14

a To the left.

b $(1, 4)$

c



15

a Downwards

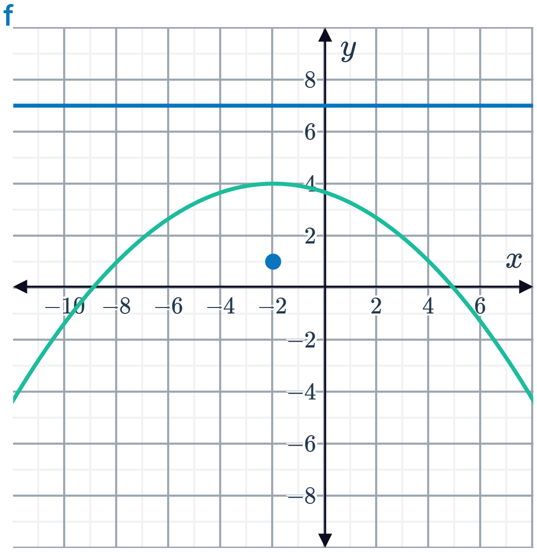
b $(-2, 4)$

c $a = 3$

d $(-2, 1)$

e $y = 7$

f



16

a Upwards

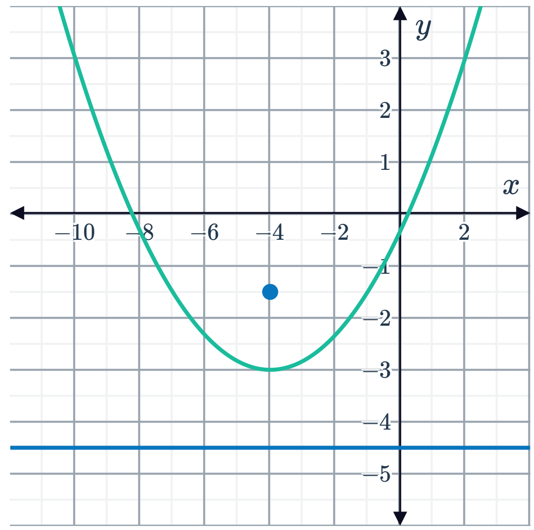
b $(-4, -3)$

c $a = \dfrac{3}{2}$

d $\left(-4, -\dfrac{3}{2}\right)$

e $y = -\dfrac{9}{2}$

f



g $\dfrac{15}{2}$ units

Equations of parabolas

17

a $x^2 = 16y$

b $x^2 = 12y$

c $y^2 = -24x$

d $y^2 = -2x$

e $x^2 = 2y$

f $x^2 = \dfrac{49}{5}y$

g $(x-6)^2 = 16y$

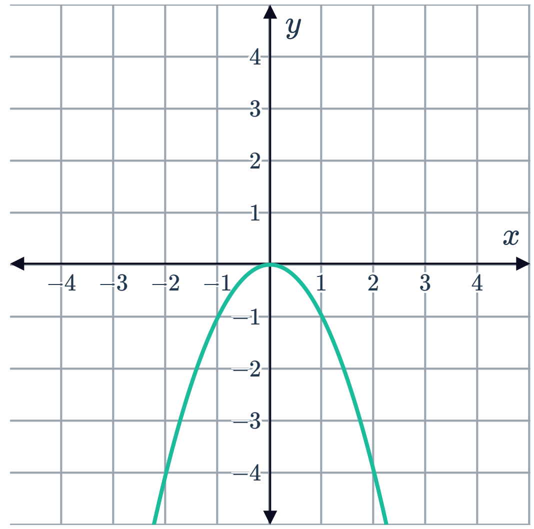
h $(y-8)^2 = 20x$

18 $y^2 = \pm 16x$

19

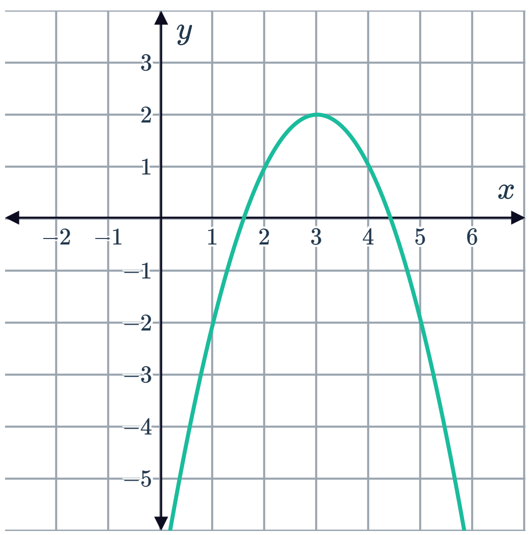
a $x^2 = -y$

b



20

a



b $(x-3)^2 = -(y-2)$

Applications

21

a $s = 12$

b 337

22

a $t = 3$

b 122